

Integrated Implementation of Five E2E Transformations in Regions (Carrier BG)

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- **Background and Benefits of Five E2E Transformations (WHY)**
- **Solutions and Regional Implementation Requirements for Five E2E Transformations (WHAT & HOW)**
- **HQ Support Resources**

Background 1: Initiate Thorough Transformation to Systematically Enhance Software Engineering Capabilities and Build Trustworthy, Quality Products

We completely redefined our R&D model with our Integrated Product Development (IPD) 1.0 transformation. Since then, we have evolved our approach to product development from one that relied heavily on individuals and chance, to one that systematically generates a continuous flow of quality products.

Moving forward, Huawei is committed to proactively embracing changes through technological innovations, and seizing new opportunities to empower the digital and intelligent transformation of the society. At the dawn of the new era, we have proposed our new vision and mission: Bring digital to every person, home, and organization for a fully connected, intelligent world.

As cloud, digitization, and software-defined everything become more and more prevalent in the digital and intelligent world, and cyber attacks and threats become the norm, demand for the trustworthiness of ICT infrastructure products is reaching entirely new heights. **Whether or not our customers want to buy our products, or even dare to, will come down to whether or not they trust them. The same goes for earning the acceptance and trust of governments the world over. Trustworthiness is critical to the survival of Huawei. Only by truly building trustworthiness can we fulfill our vision and mission in the future.**

To this end, the Board of Directors (BOD) decided to initiate thorough transformation to systematically enhance software engineering capabilities and build trustworthy, quality products.

- *Huawei BOD NT. No. [2018] 001 – Notice of Resolution on Initiating Thorough Transformation for Enhancing Software Engineering Capabilities to Build Trustworthy, Quality Products*
- *E-mail Speech No. [2019] 001 – Comprehensively Enhancing Software Engineering Capabilities and Practices to Build Trustworthy, Quality Products — An open letter to all Huawei employees*

Background 2: Implement Five E2E Transformations to Help Customers Mitigate Risks on Live Networks and Create Value for Customers

I. Why implement five E2E transformations?

1. Cyber security has become a matter of national security, with cyber attacks occurring frequently. In the wake of a series of security incidents, many countries have accelerated the promulgation of laws, regulations, and standards to address the ever-increasing challenges surrounding cyber security and data protection.
2. In the 10 years leading up to the Transformation Program for Enhancing Software Engineering Capabilities, Huawei products had constantly faced cyber security challenges, and had a lot of catching up to do in terms of vulnerability management and prevention of supply chain attacks.

II. We should truly understand what the five E2E transformations actually entail.

III. People who assume E2E responsibilities should take charge of the five E2E transformations. The organizations and owners that support the implementation of the transformations must also be clarified.

The presidents of our six business domains — as the owners of the five E2E transformations — should clearly understand what these transformations mean, why they are necessary, how they can be implemented effectively, and which support organizations should be established. In the future, **those who assume the following positions for the first time must be trained on both the five E2E transformations and the IPD system and pass related exams: president of any of the six business domains, IRB chair, product line president, IPMT director, and SPDT leader.**

Organizations supporting the implementation of the five E2E transformations: Given that these transformations span multiple business organizations and processes, their support organizations should be corporate-level and BU-level quality and operations departments, rather than organizations that work in a single process domain. The corporate Quality, Business Process & IT Dept supports company-wide processes and IT systems, so it should be responsible for driving corporate efforts to roll out the five E2E transformations and build all processes, IT systems, and organizations required for these transformations. The six business domains should establish their own five E2E transformations integration teams, and their presidents should serve as the owners, supported by their quality and operations department directors. Rep offices should establish their own five E2E transformations work groups to ensure effective implementation. **Both the owners and team leaders must be personally involved in operations and management and play a leading role.**

- *Implement Five E2E Transformations to Create Value for Customers — Minutes of the Five E2E Transformations Workshop of the Leadership Team for the Transformation Program for Enhancing Software Engineering Capabilities on August 3, 2023*

About Five E2E Transformations: Huawei Initiated the Transformation Program for Enhancing Software Engineering Capabilities in 2019, and Passed On the Transformation Requirements to Regions and Customers Following the Three E2E Transformations, Four E2E Transformations, and Five E2E Transformations Proposed in 2020, 2021, and 2022, Respectively

First proposed as three E2E transformations in 2020

- In April 2020, the Transformation Leadership Team proposed for the first time that **"product and contract lifecycle management, vulnerability management, and trustworthy release and deployment"** are three E2E transformations.

Extended to four E2E transformations in 2021

- On June 26, 2021, upon the approval of the Transformation Leadership Team, the **E2E certificate management team** was officially established. The Transformation Leadership Team also required that each product line integrate certificate management transformation into the three E2E transformations for implementation.

Changed to five E2E transformations in 2022

- On May 9, 2022, upon the approval of the Transformation Leadership Team, the **security configuration team** was established to "systematically build E2E product security configuration capabilities, demonstrate such capabilities to customers, and enable customers to achieve basic security during runtime".

E2E product and contract lifecycle management

Mitigate risks of lifecycle clause implementation in contracts and existing security risks by optimizing product lifecycle planning, building the design for lifecycle (DFLC) capabilities, and optimizing and implementing lifecycle clauses in contracts.

E2E vulnerability management

Effectively support the mitigation of vulnerability risks on customers' live networks by giving notifications about high-risk vulnerabilities within minutes and providing investigation within hours, mitigations within 3 days, and patches within 30 days.

E2E integrity protection

Ensure source trustworthiness, anti-tampering, and traceability by establishing an E2E integrity protection process and tool system, and directly delivering software packages to customers.

E2E certificate management

Eliminate risks associated with preset certificates on live networks and build certificate management as a competitive edge of Huawei's products and solutions by specifying certificate management responsibilities, optimizing certificate management rules, and developing technical solutions for certificate lifecycle management.

E2E security configuration

Systematically build E2E security configuration capabilities for products, explicitly demonstrate the capabilities to customers, and enable customers to ensure basic operating security.

Transformation objectives

Level-1 incident of a carrier

The version on the live network reached EOS but defects were not handled in a timely manner. As a result, services were interrupted for several hours and millions of users were affected.

Apache Log4j2 vulnerability

Global networks were widely affected. If carriers failed to handle the vulnerability in a timely manner with the support of suppliers, serious cyber security risks would occur.

SolarWinds incident

Hackers intruded the software tool chain, causing inherent viruses in the software.

O2/SoftBank network disconnection due to the expiration of a competitor's certificate

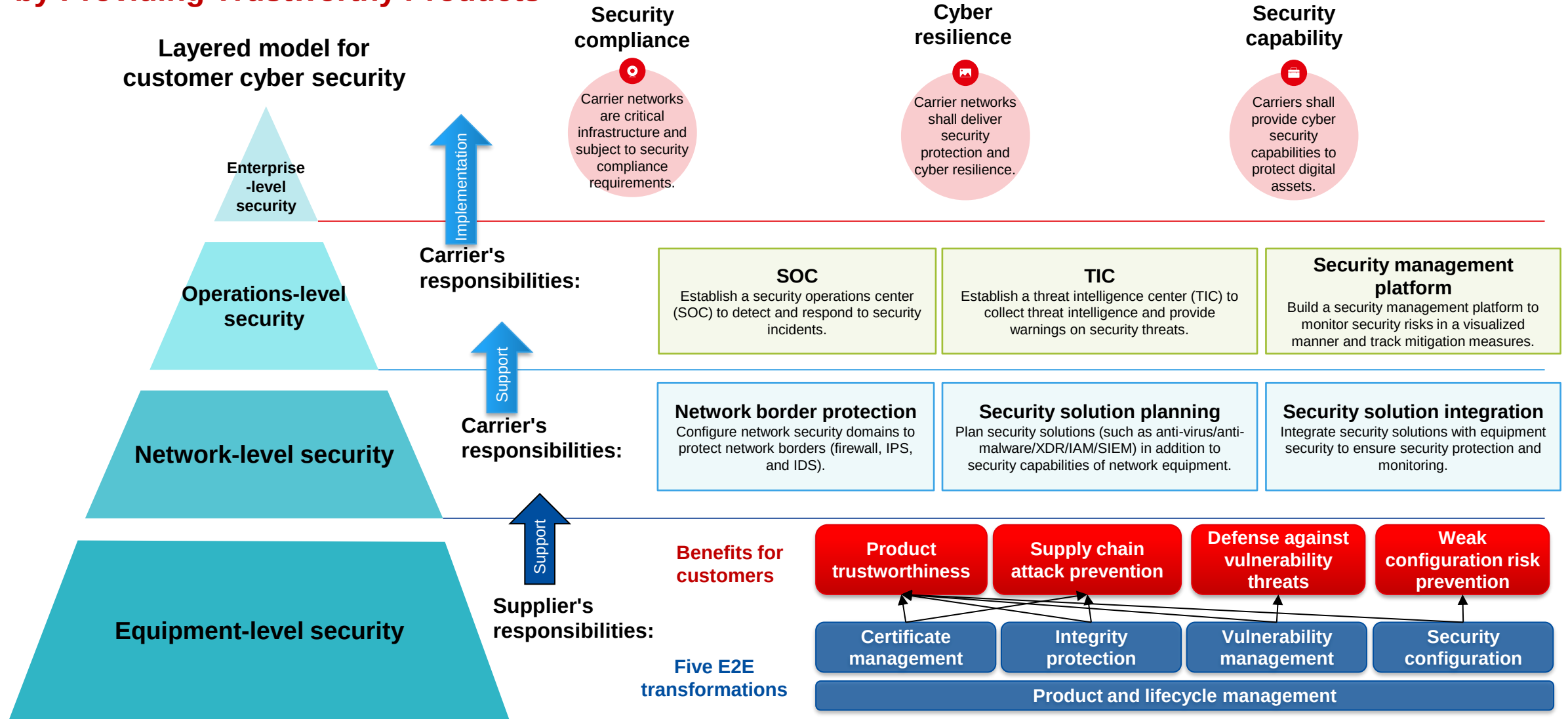
The certificate for the live network expired and was not promptly updated. As a result, NE messages became unreliable, and the entire network was interrupted for several hours.

UNC1945 intrusion by exploiting a carrier's weak configurations

Due to weak configurations, customers suffered APT attacks and all devices were subject to remote control.

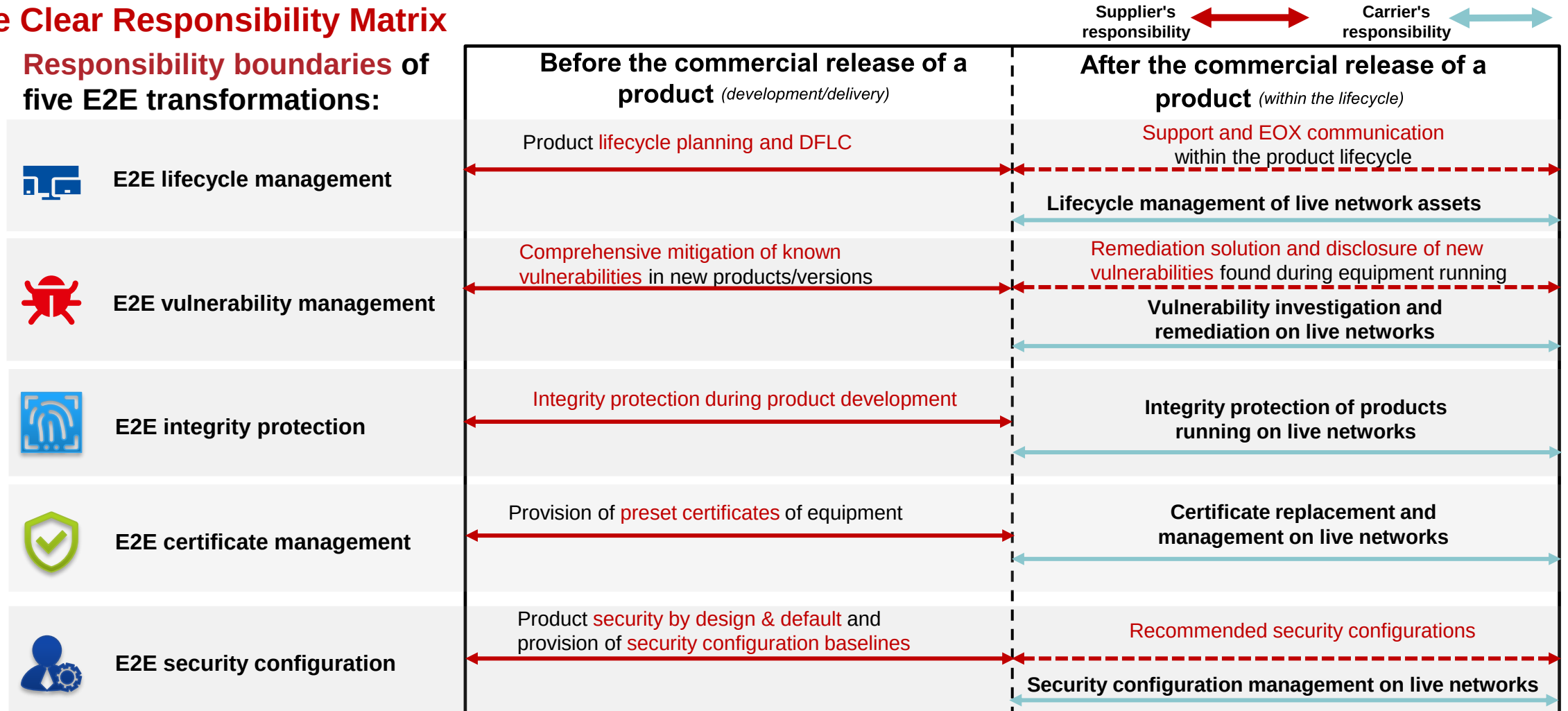
Risk examples

Benefits of Five E2E Transformations for Customers: Huawei, as a Supplier, Can Further Help Customers Implement Security Compliance, Cyber Resilience, and Security Capabilities by Providing Trustworthy Products

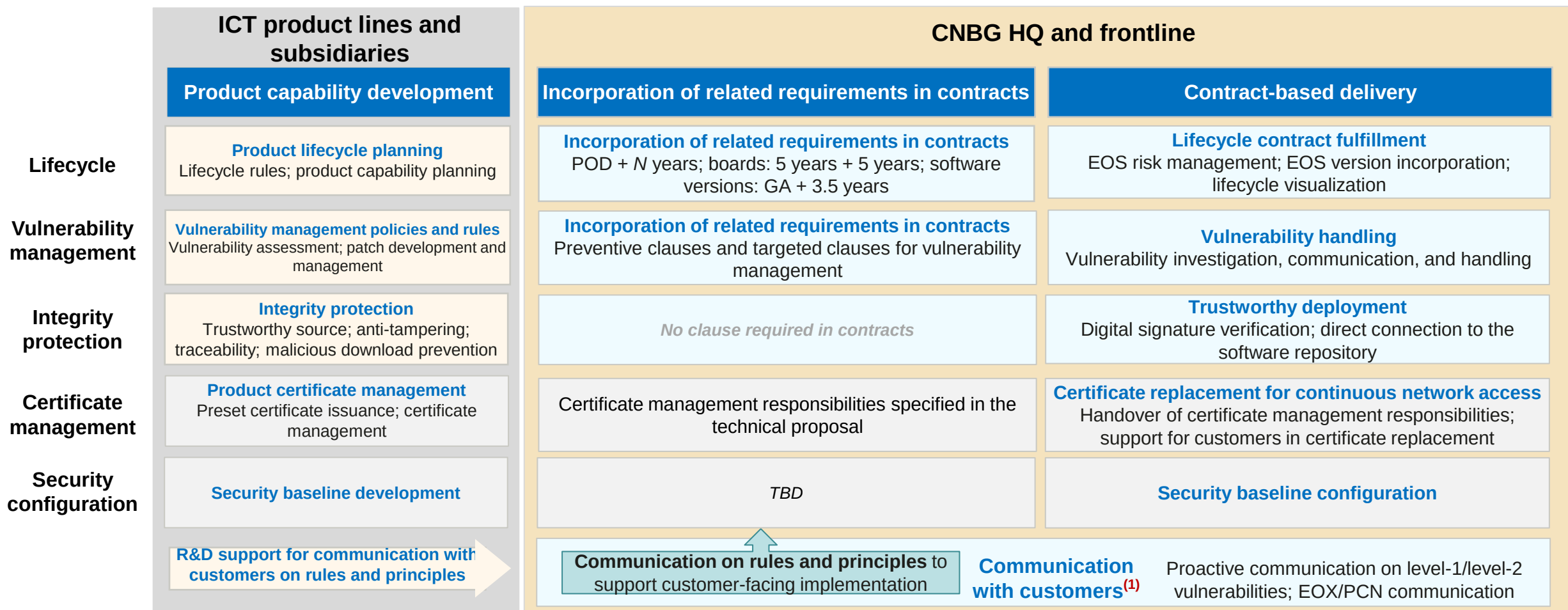


Responsibility Boundaries: Through Five E2E Transformations, Huawei Continues to Fulfill Its Responsibilities and Provide Trustworthy and Quality Products, so that Customers Want to Buy Huawei's Products, or Even Dare To. However, It Does Not Mean that Huawei Helps Customers Resolve All E2E Issues. Instead, Both Parties Must Take On Their Own Responsibilities According to the Clear Responsibility Matrix

Responsibility boundaries of five E2E transformations:



Integration of Five E2E Transformations: R&D Dept Is Responsible for Product Capabilities, CNBG for Contract Clause Implementation and Contract-based Delivery, CNBG HQ for Rules/Processes/IT Systems/Capability Support, and the Frontline for Implementation



(1) Focus: Professional communication about rules and principles with customers based on their actual situation to support the implementation of contract clauses, with support provided by R&D personnel

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Overall Requirements for Implementing Five E2E Transformations in Regions: Ensure Proper Market Insight, Customer Communication, Clause Incorporation into Contracts, Risk Management, and Contract Fulfillment

ICT Infrastructure Managing Board Document (2021)

Implementation Requirements of "Four E2E Transformations" in Rep Offices and Account Departments (security configuration was not included in 2021)

Objective

It is essential to ensure integrated implementation of these four E2E transformations on the customer interface. Whether it be arrangements for product/contract lifecycles, service level agreements (SLAs) for vulnerabilities, or the protection of software integrity (tamper resistance) during trustworthy release and deployment, customer-facing commitments always prevail. The ultimate goal of the four E2E transformations is to achieve smooth product/version launch, deployment, operation, unification, and retirement. **It is also necessary to ensure effective management of contracts and risks on live networks, as well as effective support and security assurance for products on live networks.**

Market insight

- **Analyze and develop insights into applicable national and regional laws and regulations as well as customer requirements** related to product and contract lifecycle management, vulnerability management, certificate management, integrity protection, and security configuration, and guide product teams to enable and optimize management of these elements.

Customer communication

- **Reach consensus with customers by communicating Huawei's management principles and rules on E2E transformations and aligning these principles with those of customers.**
- Understand customer requirements, develop customer communication and implementation solutions for product and contract lifecycle management, vulnerability management, certificate management, etc., and support the creation and fulfillment of contract clauses.

Incorporating clauses on E2E management responsibilities and obligations into contracts

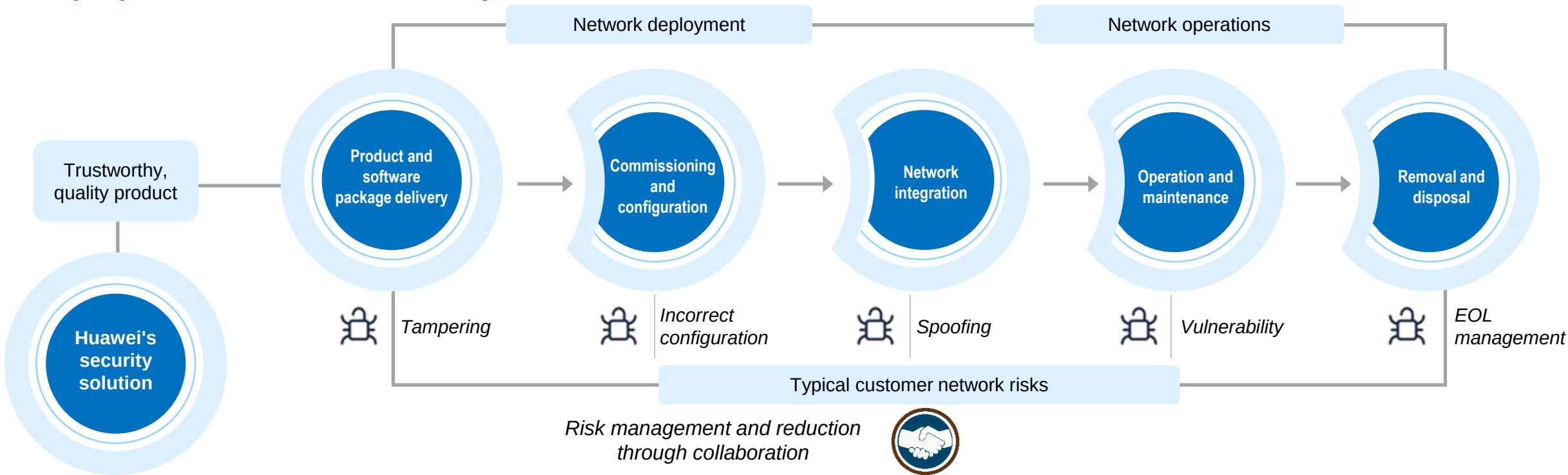
- Incorporate lifecycle-related clauses into new contracts, and include preventative clauses related to vulnerability management, **in order to maintain Huawei's required scope of responsibilities therein.**

Risk management on live networks

- **Continuously manage the installed base by modernizing legacy networks, replacing products and boards that have passed EOX, and upgrading and unifying outdated versions, in order to mitigate the risks of outdated products and software.**
- Effectively manage risks and communicate with customers in advance for products, boards, or software versions that are approaching the end of their service lifespan or EOS, and **urge customers to replace or upgrade the products, boards, or software versions in question.**
- In principle, do not sell warranty services for products that have reached the end of their service lifespan or EOS. In special scenarios where such services can be sold, **submit an application for review and decision through the exceptional service sales process, and implement the final decisions.**
- At the service delivery and maintenance stages, support the mitigation of risks on customers' live networks by adhering to the delivery scope stipulated in the contracts.

For detailed requirements, refer to: <https://w3.huawei.com/info/en/doc/viewDoc.do?did=5067693&cata=184183>

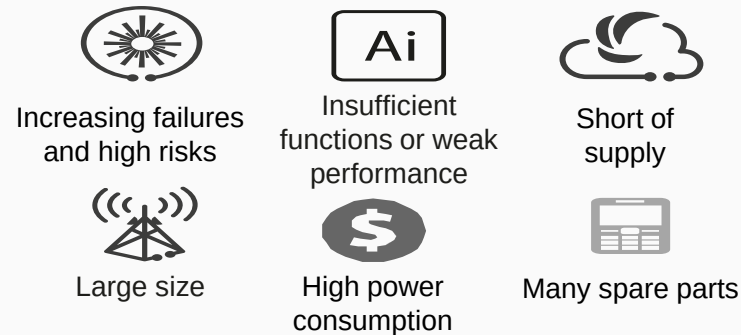
Huawei's E2E Security Solution: Provide Quality Products and Security Capabilities from the Perspectives of Five E2E Transformations to Reduce Risks Faced by Customers from Network Deployment to Live Network Operations



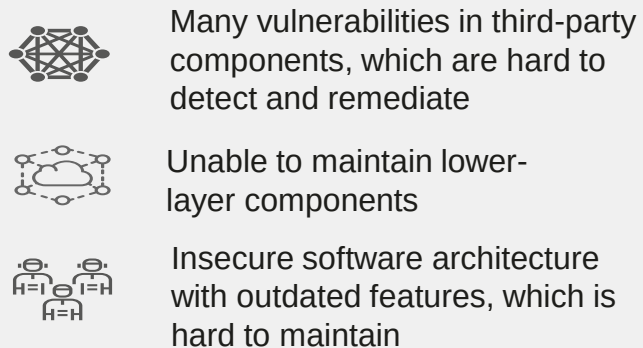
Typical Risk	Security Solution	Description
Tampering	Integrity protection	Digital signature; Huawei Online Upgrade Repository (HOUP)
Incorrect configuration	Security configuration	Security configuration baseline; configuration check tool
Spoofing	Digital certificate management	Certificate deployment and operations solution
Vulnerability	Vulnerability mitigation	PSIRT; vulnerability management mechanism; multiple vulnerability disclosure channels
EOP management	Lifecycle management	"POD + N years" mechanism; EOX notification

Why Lifecycle Management Is Required: Improve Reliability and Security to Avoid Major Quality Risks

Legacy hardware



Legacy software



Evolution of communications standards and new services and technologies

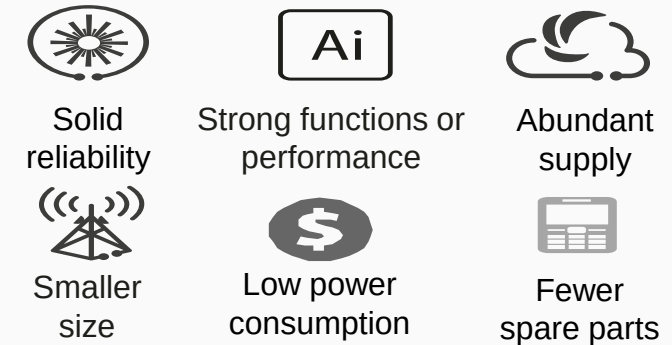
Device aging

OPEX reduction

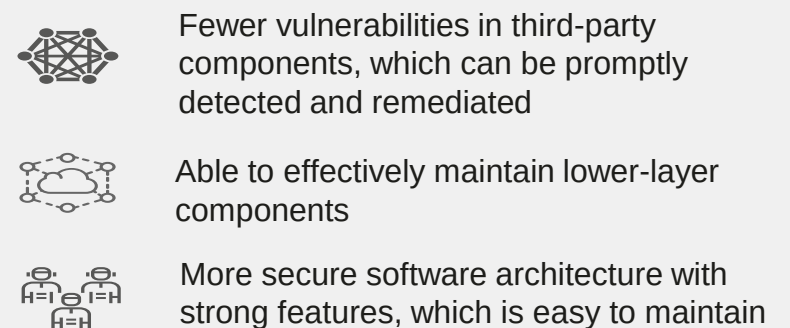
Software architecture evolution and third-party software update requirements

Carriers' asset management requirements

New hardware



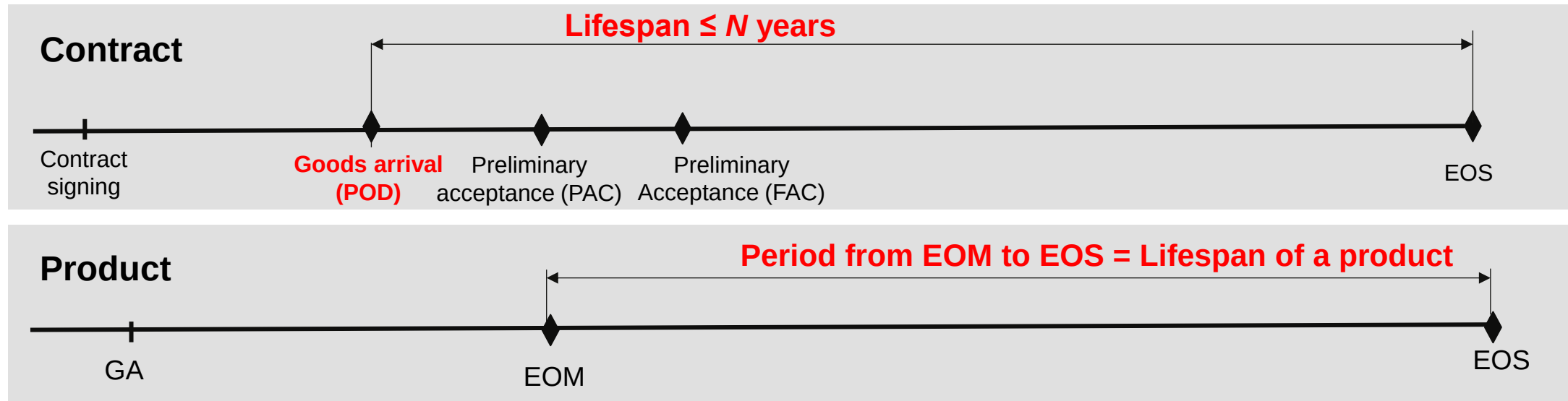
New software



Contract Clauses: Ensure that the Lifespan Committed in a Contract Does Not Exceed the Planned Product Lifecycle

The frontline personnel shall sign customer contracts based on the basic lifecycle requirements of each domain.

- ① The lifespan committed in a contract starts from the date of goods arrival (POD).
- ② **Committed product lifespan (N years):** It shall not exceed the EOS date of the product or the sum of the POD date and the committed lifespan baseline. In addition, disclaimers shall be provided to specify that Huawei is exempted from liabilities after $\text{POD} + N$ years.

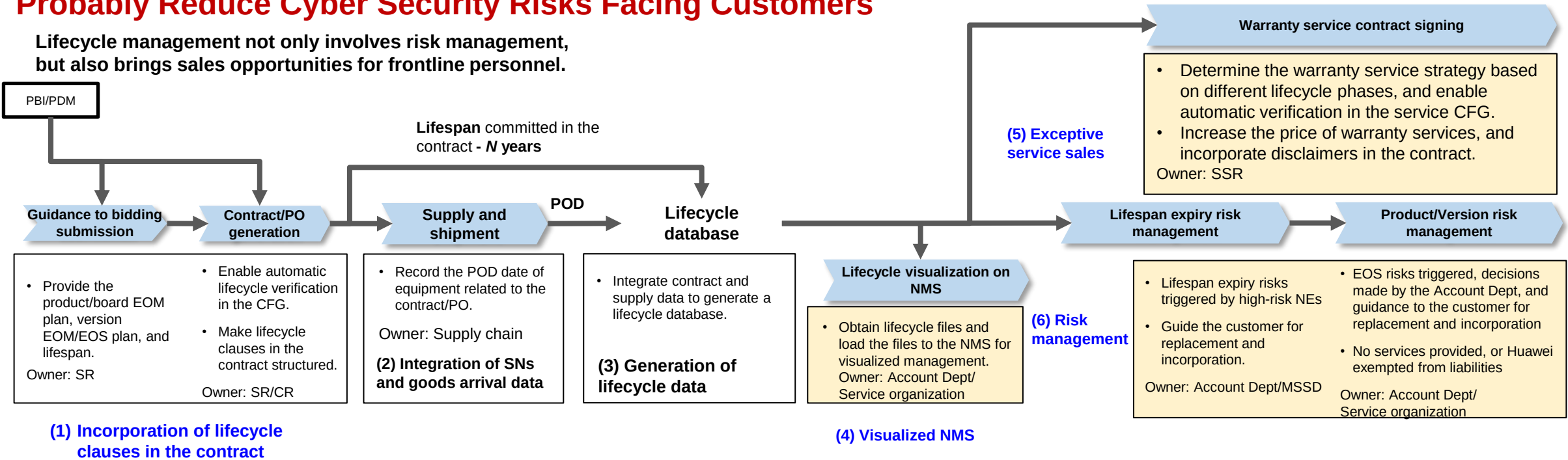


Product lines shall plan the product lifecycle to meet the basic lifecycle requirements of each domain.

- ① **Product lifecycle planning (period from EOM to EOS):** It shall be longer than or equal to the committed lifespan baseline.
- ② It shall be implemented in activities such as selection of devices and software components, product and board design, and inter-generational planning of products and boards.

Product and Contract Lifecycle Management: Lifecycle Management Is the Basis of Equipment Cyber Security Risk Management; Routine Management and Orderly Iteration Can Probably Reduce Cyber Security Risks Facing Customers

Lifecycle management not only involves risk management, but also brings sales opportunities for frontline personnel.



Sales (LTC): Incorporate lifecycle clauses in the contract and accurately enter related information.

- New contracts:** Incorporate lifecycle baseline clauses in all new contracts.
- Existing contracts:** If risk clauses are involved in existing contracts, properly manage risks and optimize lifecycle clauses when appropriate.
- Contract entry:** Ensure that the entered product lifecycle information is correct.

Engineering delivery (SD)

- Ensure visualized product lifecycle information (on the NMS).
- Lay a foundation for subsequent warranty service sales, version incorporation, and product replacement through information visualization.

O&M (SD)

- Risk management:** Routinely manage lifecycle risks of products, boards, and software versions, and remind customers of product replacement and version incorporation in advance.
- Exceptional sales of warranty services beyond the lifecycle:** Increase the price of warranty services, and incorporate disclaimers in the contract.
- Notify customers before lifecycle expiry.

Vulnerability Characteristics: Harm, Sensitivity, and Emergency

Three characteristics of vulnerabilities:

Harm

Sensitivity

Emergency



Heartbleed in 2014 showed us the harm of vulnerabilities.

WannaCry in 2017 showed us the sensitivity of vulnerabilities.

Log4Shell in 2021 showed us the emergency of vulnerabilities.

Harm: Heartbleed



Impact

More than 17% of global encryption service websites (more than 1 million websites) were affected, resulting in a direct loss of more than USD500 million.

“One of the most consequential since the advent of the **commercial** Internet

Takeaway

If a **vulnerability** is generated and exploited by hackers, **it will cause great harm to asset owners**, including identity theft, data theft, and communication eavesdropping.

Sensitivity: WannaCry



About 200,000 computers in more than 150 countries were attacked, causing a global loss of over USD4 billion.

“Biggest **cyber attack** around the world **affecting numerous users**

Vulnerabilities can also be exploited by national actors to **make offensive cyber weapons** or **launch large-scale cyber attacks**.

Emergency: Log4Shell



More than 90% of global Java programs and more than 93% of enterprise cloud services were affected.

“One of the most serious **vulnerabilities ever**: Easy to be exploited, serious, and ubiquitous

Vulnerabilities bring great challenges for vendors, customers, and users. **They have to take quick actions before vulnerabilities are exploited by hackers**, and **may be held accountable** by regulators.

Core Values of Vulnerability Management: Ensure Compliant Product Admission, Meet Customer Requirements, Help Customers Mitigate Risks, Build Trust, and Improve Security Competitiveness



Compliance

As vulnerability management gradually evolves into a legal or regulatory requirement of countries, it is a part of compliance.



Customer requirement

In the face of increasing cyber risks and compliance pressure, customers may impose contractual requirements on equipment vendors for more comprehensive vulnerability management.



Risk management

Improper vulnerability management may lead to cyber security incidents for customers and cyber security crises for Huawei. Proactively managing product and service vulnerabilities is an important measure to mitigate Huawei's risks and help customers mitigate cyber security risks on live networks.



Boosting confidence

Vulnerability management builds customers' confidence in Huawei's security capabilities in the cyber security domain and provides continuous testament to our cyber security capabilities.



Security competitiveness

The capabilities to keep potential vulnerabilities at low risks, get fully aware of known vulnerabilities, promptly fix them, assess risks, and provide easy-to-deploy patches are the core cyber security competitiveness from the perspective of customers.

Huawei's Vulnerability Management: 1 Objective, 5 Principles, 3 Phases, and 5 Key Activities



Vulnerability Management Requirements: Align Understanding with Customers, Establish a Routine Management Mechanism, and Cooperate to Manage Vulnerability Risks

	Pre-sales	Sales	Delivery	Maintenance		Routine			
	Understanding alignment	Clause incorporation	Patch installation	Communication on high-risk vulnerabilities	Transfer of patch information via RNs	Vulnerability risk mitigation	CERT-to-CERT connection (disclosure website)	Customer issue handling	Insights into customer requirements
Incorporation into processes	MCR	LTC	SD	MCR & SD	SD	SD	MCR	ITR	MCR & RM
Key requirement	- Vulnerability management requires cooperation between carriers and vendors. - Vulnerabilities are inevitable but can be managed. - Ensure hierarchical vulnerability management.	Scenarios where customers have no demands: Incorporate preventative clauses related to vulnerability management in all newly signed contracts. Scenarios where customers have demands: Involve product lines in the review/decision-making of targeted clauses related to vulnerability management. - Properly manage risk clauses in existing contracts.	- Install the latest patches in the engineering delivery phase. - Check that the latest patches are installed before a project is transferred to maintenance.	- Proactively and promptly notify customers of high-risk vulnerabilities. - The Account Dept should specify the communication strategy (customer level, timing, and key messages). - Communication with O&M customers is mandatory, and communication with security customers and C-level executives is optional.	Understand customers' vulnerability management habits and processes, and in line with the "one customer, one policy" principle, regularly communicate with customers via RNs to transfer information about remediated vulnerabilities.	Fulfill responsibilities based on contracts.	- Provide it for carriers with vulnerability management capabilities and demands. - Account Depts shall take charge of customers' demands for CERT-to-CERT connection. - Perform routine operations and management after CERT-to-CERT connection (e.g., customer information management).	Handle and resolve vulnerabilities through iCare tickets.	- Gain insights into customers' vulnerability management requirements. - Provide customers' feedback on vulnerability management.
Key role	All	Commercial manager and solution manager	Service engineer	Service engineer/Security professional/CC3	Service engineer	Service engineer	CC3/Security professional	Service engineer	CC3/Security professional/Service engineer
Implementation frequency	As required	New contract-triggered	New project-triggered	Event-triggered	Quarterly (recommended); at least semi-annually	Based on contracts	As required	As required	Regularly

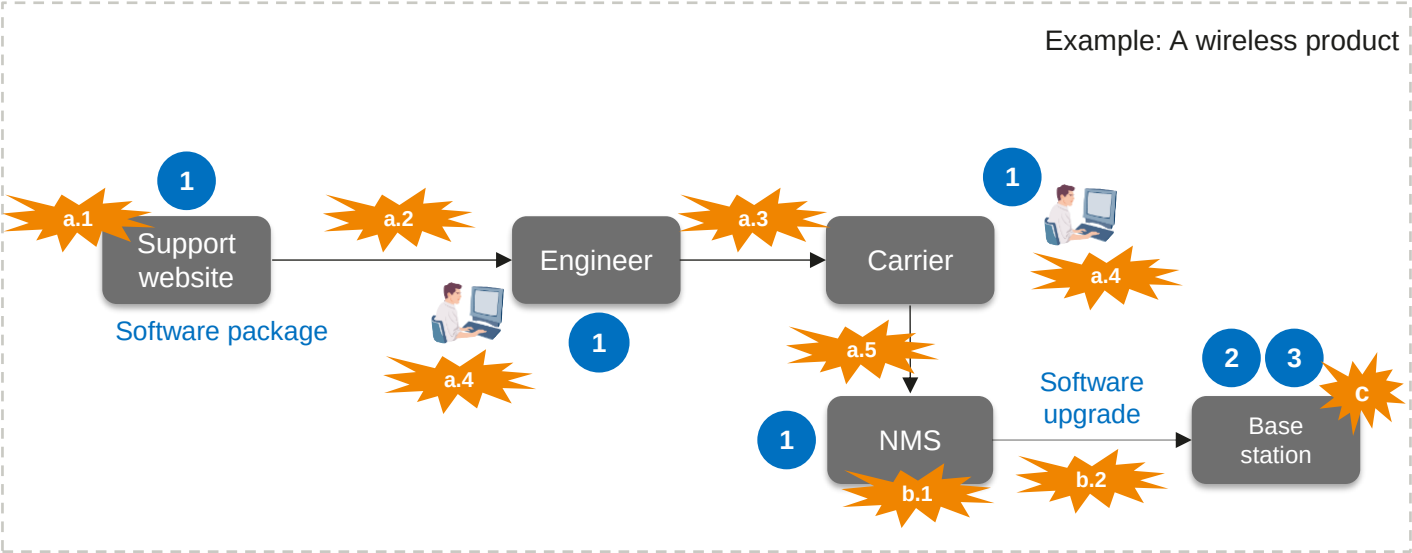
Note: Security professionals refer to KAD security contact persons, CSPOs, RSPOs, and others engaged in security work.

Frequent Attacks on the Software Supply Chain Have Made Effective Prevention a New Focus of Suppliers and Carriers



Gartner predicts that by 2025, 45% of organizations worldwide will have experienced attacks on their software supply chains, a three-fold increase from 2021.

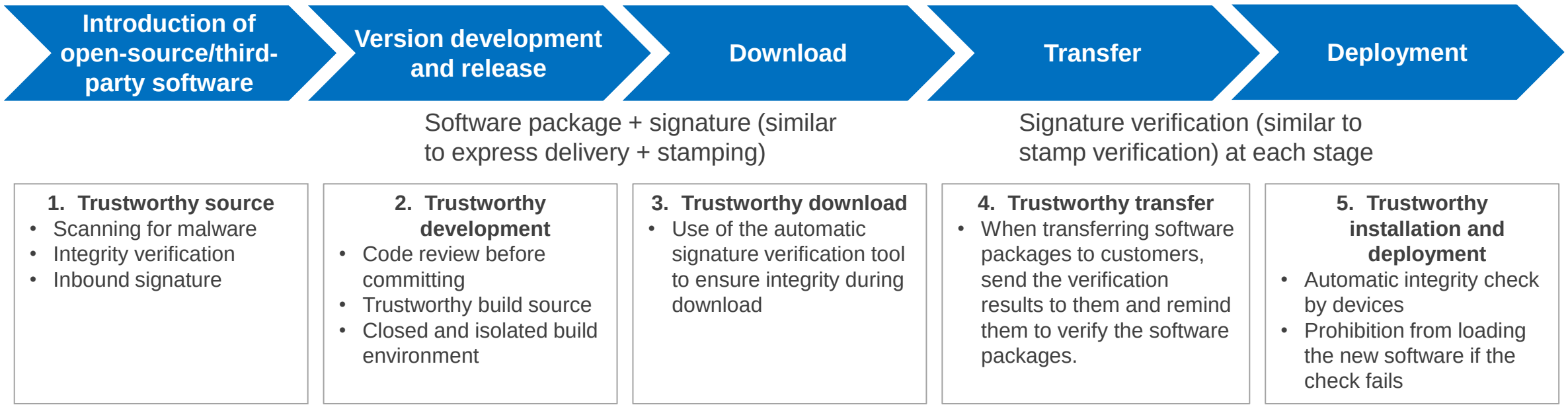
<https://www.gartner.com/en/articles/7-top-trends-in-cybersecurity-for-2022>



Potential Attack Target	Technology	Application Scenario
a.1 Software download website (Support website) a.2 Download channel: Support website -> Huawei engineer a.3 Transfer channel: Huawei engineer -> Carrier engineer a.4 PC, server, and engineer a.5 Upload channel: Carrier engineer's PC/Server -> NMS	Digital signature	Software transfer phase
b.1 NMS b.2 Download channel: NMS -> Base station	Digital signature	Software installation phase
c. Base station (e.g. exploits and malware)	Secure boot	Software running phase

Objectives of Integrity Protection: Prevent Tampering/Implantation/Malware/Spoofing in an E2E Manner Through Trustworthy Source, Internal R&D Activities, and External Delivery

Integrity Protection



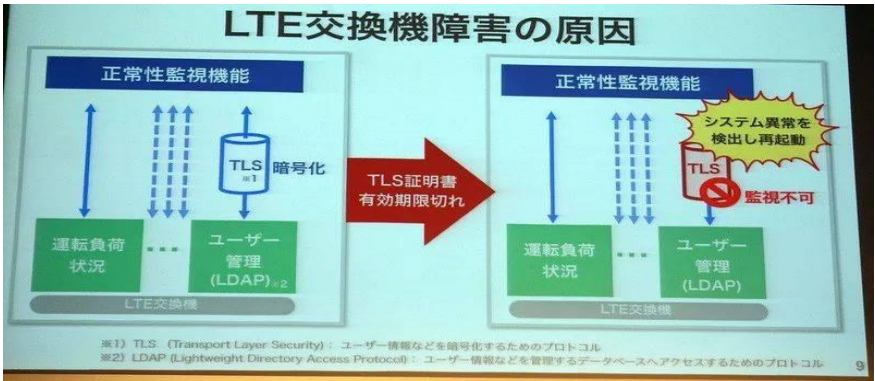
Delivery scenarios

- Delivery by Huawei: Use ICS Lite to download software and perform automatic signature verification.
- Delivery by customers: When providing software packages to customers, send the integrity verification results to them and advise them to check the software.

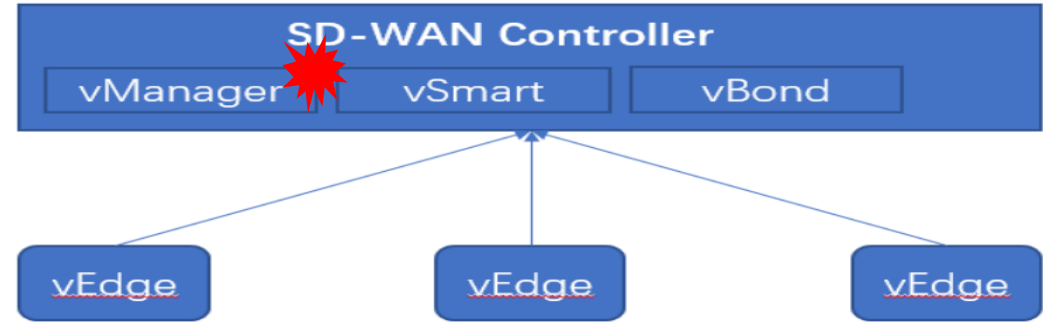
Multiple Incidents Occurred in the Industry Due to Improper Certificate Management and Certificate Expiration

- Each certificate has a lifecycle. If preset certificates are not replaced, certificate leakage may occur; if certificates expire, customer service failures may occur.
- Risks of Huawei's existing preset certificates on the live network:** A small number of preset certificates will expire before 2027, and **tens of millions of preset certificates will expire since 2027. (If the certificates are not properly managed, service failures will occur.)**

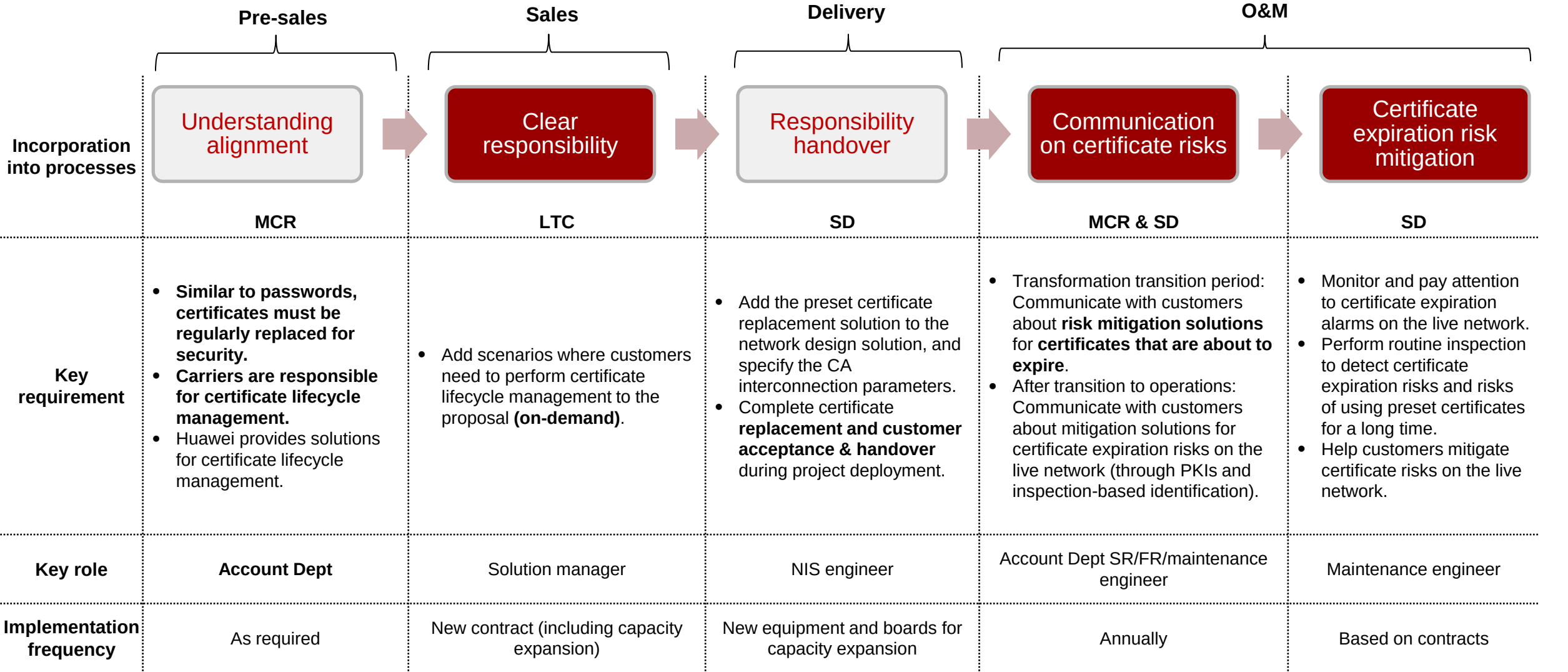
Incident 1: In the afternoon of December 6, 2018, **an exception occurred after Vendor E's certificate expired**. 18 4G core network NEs of Vendor E in the central equipment room of Carrier S in Country J became faulty, causing network-wide communication failures among a large number of users. The fault **lasted for 4 hours and 25 minutes**. As a result, about **30.6 million users** failed to communicate with each other.



Incident 2: According to foreign media reports on May 11, 2023, Vendor C's **SD-WAN network was interrupted due to CPE certificate expiration**. SD-WAN users said that the downtime **had a great potential impact on enterprises worldwide**. (Vendor C rated the severity of this incident as "Level 1 - Catastrophic" on its official website.)



Five Steps for Regional Implementation: Align Understanding with Customers, Prevent New Risks During Delivery, and Eliminate Risks of Existing Certificates During Maintenance



Personal Information of More Than 14 Million Users of Company V Was Leaked Due to Improper Security Configurations (Weak Password)

Background:

In 2017, the personal information of more than 14 million users of Company V was **leaked due to improper security configurations of a third-party cloud server**.

After analysis, the Cyber Risk Team of UpGuard, a cyber security company, concluded that the data of as many as 14 million users was leaked due to a **misconfigured cloud-based file repository**. The cloud server was owned and operated by a telephonic software and data company, a third-party vendor for Company V.

Attack process and result

Cyber Risk Research of UpGuard discovered a cloud-based Amazon S3 data repository that was **fully downloadable and configured to allow public access**. The database and its many terabytes of contents could thus be accessed simply by entering the S3 URL.



The critical data repository in question was not owned by the enterprise, but operated by Company N, a third-party vendor to the enterprise. The third-party vendor kept the bucket publicly accessible, leading to the leakage of personal information of a large number of customers.



Finally, Company N and Company V confirmed the information leakage incident.



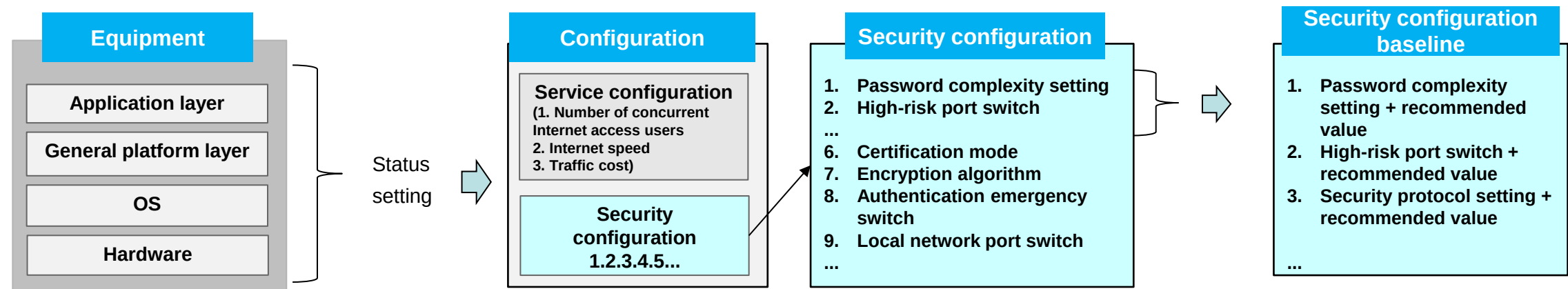
Takeaway: Weak passwords and weak configurations would allow amateur hackers to intrude into services, causing risks such as leakage of critical and personal data.

Security Configuration — Status Setting of Equipment Security Features/Functions

In general, security configuration refers to the setting of equipment security status, for example:

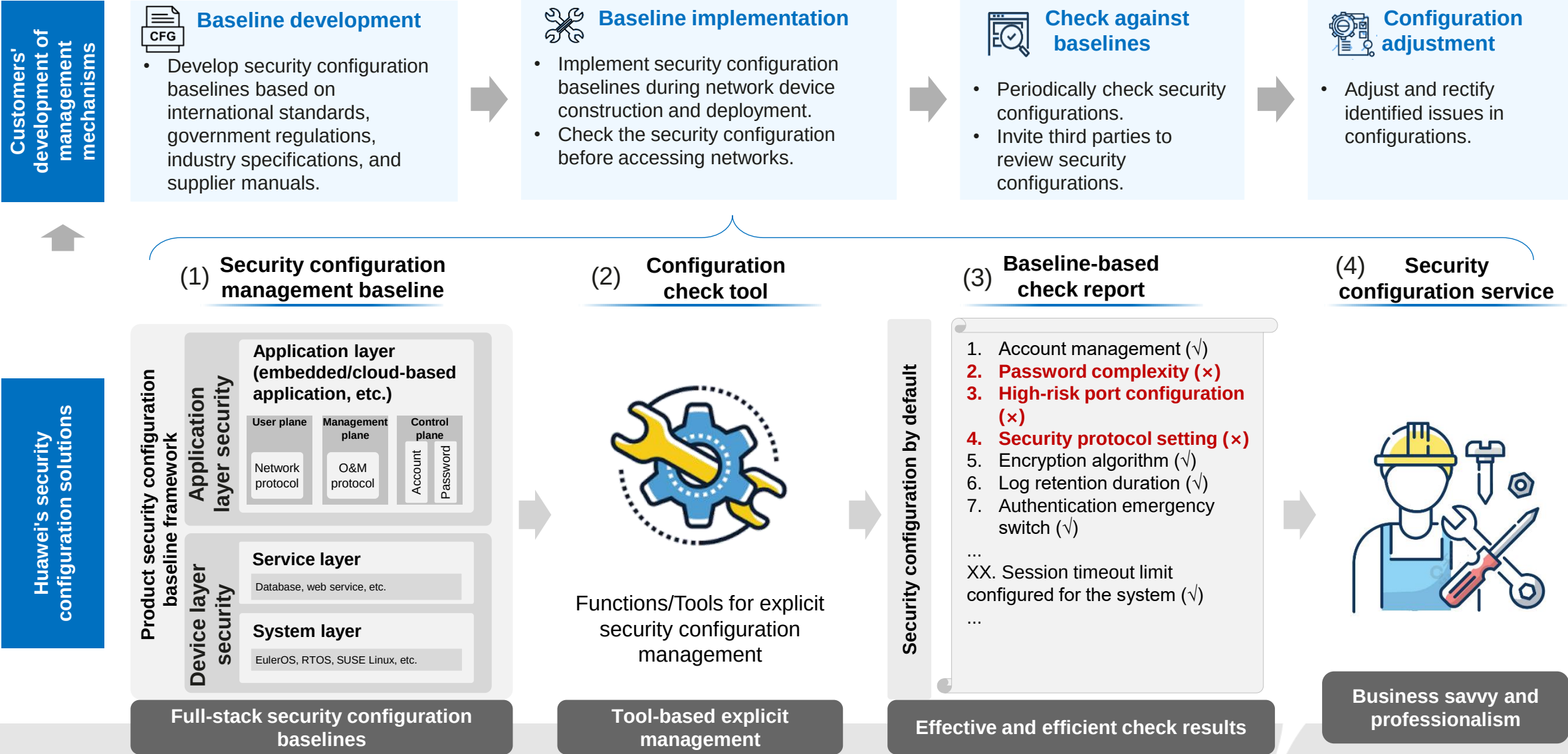
1. "Security configuration" of a house: (1) Whether the smoke detector function is enabled. (2) Door opening mode; whether a fingerprint, key, or facial recognition is required
2. "Security configuration" of the Windows OS: (1) Whether disk data encryption is enabled. (2) Whether the screen saver timeout is set, for example, 10 minutes or 5 minutes. (3) Whether a password is required to log in to the system

Security configuration concepts of communication equipment



Concept	Definition
Security configuration	Security configuration is the status setting of product security features/functions.
Security configuration baseline	A security configuration baseline is the configuration items that equipment users need to focus on and the corresponding recommended values. It is used to ensure the basic security of equipment.

Security Configuration Management Solution: Huawei Provides Basic Security Configuration Capabilities to Help Customers Manage Live-Network Security Configuration



Operating Map: Develop Regional Operating Maps for Five E2E Transformations Through Three Key Steps and Properly Allocate Resources to Effectively Manage Risks and Present Positive Value

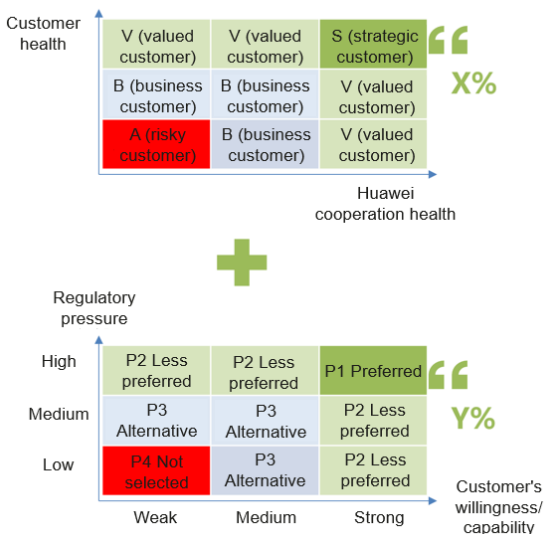
Operating intent: (1) Clarify responsibilities and mitigate risks. (2) Gain customer trust based on "process loyalty" through deep cooperation.



Gain a full understanding of the environments from three dimensions based on two operating scenarios.

Operating Scenario	Evaluation Item
1. Risk management 2. Value presentation	1. Laws and regulations; regulatory pressure 2. Customers' security maturity (four dimensions) - Security organization - Security strategy - Security budget - Security demand 3. Business situation (based on S/V/B/A classification)

Complete modeling based on key elements and classify the environments (objects).



Properly allocate resources to the environments based on priority.

Resource allocation strategy:

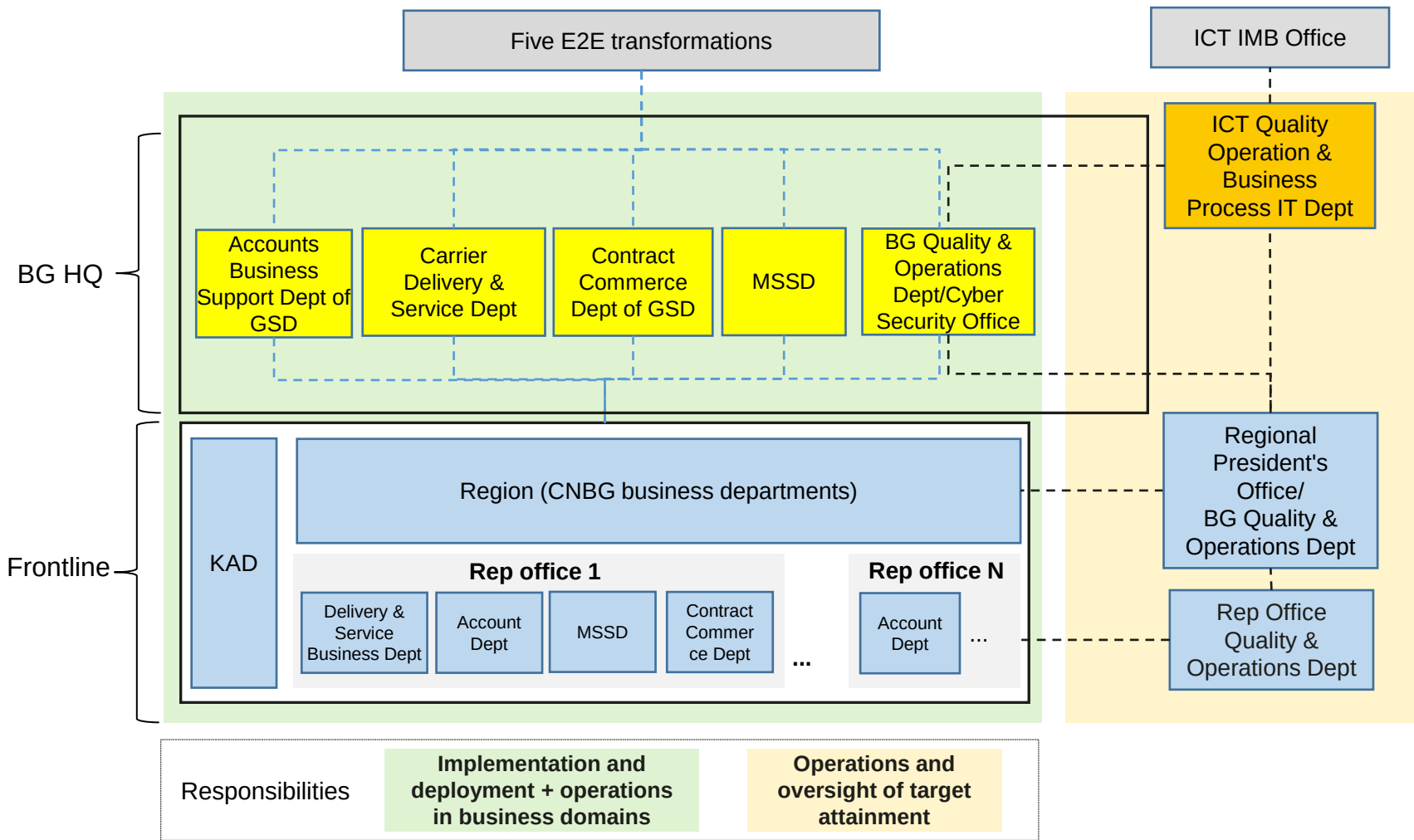
Customer Category	Operating Scenario	Resource Allocation
Preferred (with capabilities/demands)	Risk management + value presentation	Product line + BG + region
Less preferred (with laws and regulations/demands)	Risk management + value presentation	BG + region
Alternative (others)	Risk management	Region
Not selected	Risk management	Region

Result: (1) Effective risk management. (2) Positive value presentation (higher customer loyalty and customer satisfaction through organization/process/IT interconnection)

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Responsibilities of HQ Departments and Regional Organizations



Role	Responsibility
E2E transformation project team	Develop implementation capabilities, including making rules, optimizing processes, developing operation platforms and equipment, and ensuring that the conditions for operations are met.
BG business department	Manage the implementation targets of their own business domains , provide support and enablement, and build capabilities and mechanisms for implementation and operations.
BG Quality & Operations Dept/Cyber Security Team	The BG Cyber Security Team acts as a COE. The BG Quality & Operations Dept oversees the attainment of implementation targets and performs process compliance check during operations.
Regional president, rep office GM, BG director, and KAD director	Serve as the owners for the implementation and operations management of the five E2E transformations in regions , set implementation targets, and take responsibility for the final implementation results.
Regional/Rep office BG and subordinate team	Implement transformations , conduct routine operations, address issues during the implementation, and continuously promote the optimization of rules, processes, and IT systems.
Quality & operations organization at each level or designated role	Oversee the attainment of implementation targets in their own domains , push for the resolution of issues arising from implementation, and establish a continuous operations mechanism.
RSPO/CSPO	Provide professional cyber security capabilities to support security implementation (such as understanding alignment in vulnerability management) in E2E transformations and assist the director of the Quality & Operations Dept in overseeing the attainment of implementation targets.

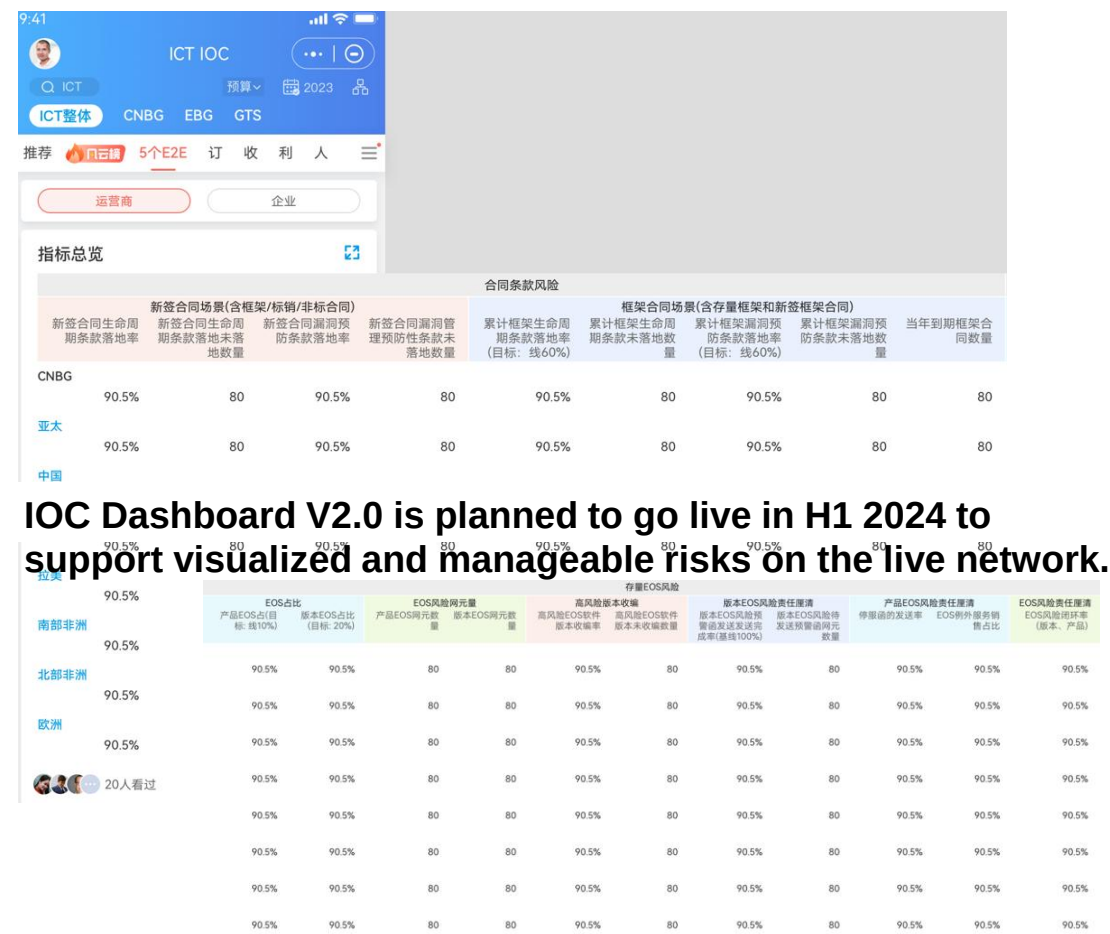
ICT document — Resolution on the Continuous Implementation Responsibilities of the "Five E2E Transformations":
<https://w3.huawei.com/info/en/doc/viewDoc.do?did=5237373&cata=184213>

Operations Platform: The IOC Dashboard on the Mobile Phone Has Gone Live, and the IOC Dashboard on the PC Is Planned to Go Live in Q1 2024

Cooperate with the five E2E transformations project team to support the HQ and regions in operations and risk management through the IOC Dashboard.



IOC Dashboard V1.0 has gone live to support visualized strategic objectives.



IOC Dashboard V2.0 is planned to go live in H1 2024 to support visualized and manageable risks on the live network.

Operations Metrics: Completion Rate + Quantity; Inventory + Incremental; Displaying Results + Supporting Process Management

- Operations Metric System V1.0:** The carrier IOC Dashboard (mobile) went live in September 2023, and all appraisal metrics can be viewed.
- Operations Metric System V2.0:** V2.0 is planned to go live in H1 2024, which will optimize the metric system based on contract clause risks, product lifecycle risks on the live network, vulnerability risks, preset certificate risks, and more.

Business Scenario	Sub-scenario	Carrier Business Operations Metric	Metric Status
Contract clause risk management	New contracts signed in the year	Percentage of new contracts with lifecycle clauses	Available in V1.0
		Number of new contracts without lifecycle clauses	To be available in V2.0
		Percentage of new contracts with vulnerability prevention clauses	Available in V1.0
		Number of new contracts without vulnerability prevention clauses	To be available in V2.0
	Historical frame contracts	Percentage of historical frame contracts with lifecycle clauses	Available in V1.0
		Number of historical frame contracts without lifecycle clauses	To be available in V2.0
		Percentage of historical frame contracts with vulnerability prevention clauses	Available in V1.0
		Number of historical frame contracts without vulnerability prevention clauses	To be available in V2.0

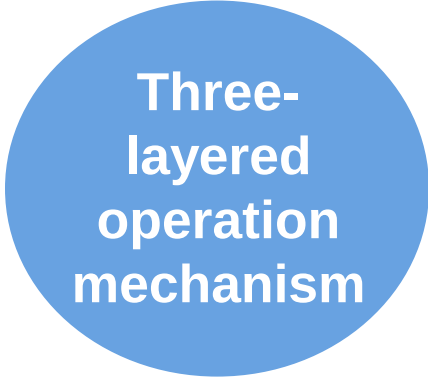
Business Scenario	Sub-scenario	Carrier Business Operations Metric	Metric Status
Software integrity protection	Tool-based software package download	Completion rate of automatic signature verification for downloaded software packages	To be available in V2.0
Certificate risk management	Mitigation of risks brought by preset certificates	Total number of expired certificates in two years	To be available in V2.0
		Risk mitigation rate of preset certificates on devices	Available in V1.0
		Number of preset certificates with to-be-mitigated risks in two years	To be available in V2.0
Security configuration	Security configuration risks	Implementation rate of mandatory security configuration items in new projects	To be available in V2.0
		Inspection: Inspection completion rate for mandatory security configuration items	To be available in V2.0
		Inspection: Security configuration risk rate of live-network NEs	To be available in V2.0

Business Scenario	Sub-scenario	Carrier Business Operations Metric	Metric Status
Lifecycle management	Product and version EOS status	Percentage of EOS products	To be available in V2.0
		Percentage of EOS versions	To be available in V2.0
		Percentage of NEs with EOS products	To be available in V2.0
		Percentage of NEs with EOS versions	To be available in V2.0
		Percentage of NEs with EOS boards	To be available in V2.0
	Incorporation of high-risk EOS versions	Incorporation rate of high-risk EOS software versions	Available in V1.0
		Number of high-risk EOS software versions not incorporated	To be available in V2.0
	Specifying responsibilities for version EOS risks	Percentage of sending version EOS risk warning letters	To be available in V2.0
		Number of NEs for which version EOS risk warning letters are to be sent	To be available in V2.0
	Specifying responsibilities for product EOS risks	Percentage of sending EOS letters	To be available in V2.0
Vulnerability risk management	Specifying responsibilities for EOS risks	EOS risk closure rate (version and product)	Available in V1.0
	Sales of maintenance services beyond the service lifespan	/	To be available in V2.0
	Exceptive sales after EOS is reached	/	To be available in V2.0
	Communication on and mitigation of high-risk vulnerabilities	Completion rate of communication on high-risk vulnerabilities	Available in V1.0
		Number of high-risk vulnerabilities that have not been communicated	To be available in V2.0
		Completion rate of patch installation on NEs with high-risk vulnerabilities	To be available in V2.0
		Number of NEs with high-risk vulnerabilities that have not been patched	To be available in V2.0
		RN-based vulnerability information communication rate	To be available in V2.0

Operations Mechanism: Build a Three-layered Operation Mechanism Consisting of the Joint Q&O Meeting, ST Meeting, and 3T Meeting to Guarantee Daily Operations and Management

1. Rep office **joint Q&O meeting**, for daily operations and management of the five E2E transformations

- Owner: **Director of the Quality & Operations Dept or GMO**
- Attendees: Rep office personnel and PCs
- Objectives:
 1. Take charge of **daily operations and management**, and identify and resolve risks and issues found during operations.
 2. Based on the business practices of the rep office, **identify the scenarios (L5/L6) where process adaptation is required** in the rep office.



Three-layered operation mechanism

3. CNBG/EBG/GTS/MSSD **3T meeting**, for decision-making on process optimization

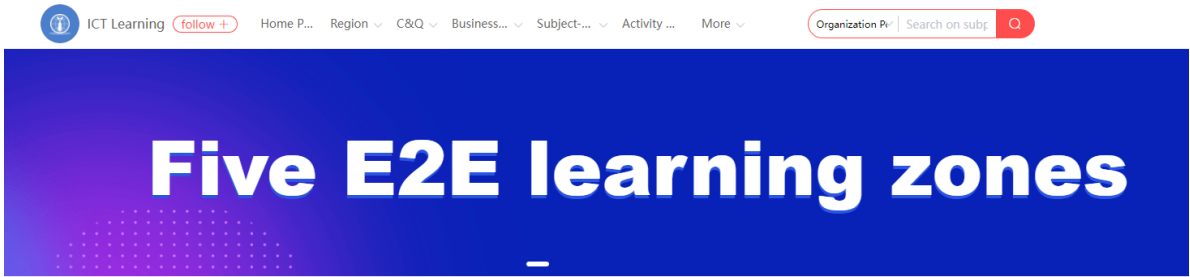
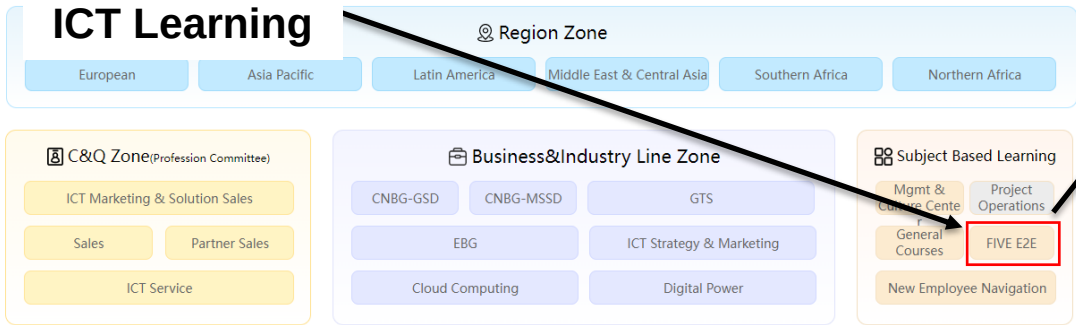
- Owner: **CNBG/EBG/GTS/MSSD GPO**
- Attendees: **BPOs**
- Objectives:
 1. **For new solutions**, incorporate them into the regional deployment and operations planning.
 2. **For common issues**, optimize business rules and processes/IT systems after decision-making at the 3T meeting.

2. Rep office **ST meeting**, for plan and progress review and resolution of key issues

- Owner: **GM of the rep office**
- Attendees: Directors of account departments and business departments
- Objectives:
 1. The Quality & Operations Dept at each level or GMO is responsible for organizing the meeting. The organization's annual plan on the five E2E transformation implementation and operations will be reviewed and released at the ST meeting in January each year.
 2. Regularly review the execution progress, coordinate resources, and resolve issues found during operations.

Enablement Platform: One-Stop Learning Zone, Facilitating Self-learning and Examination in Regions

- **Self-learning:** Frontline personnel can select learning content based on roles and topics.
- **Regional experience sharing:** Excellent experience and practices of regions are collected to encourage mutual learning.
- **One-stop information acquisition:** Documents, guides, and FAQs on the five E2E transformations are archived in a unified manner.



Training courses

Non-specialist positions	Country General Manager	Carrier Business Dept Director/Account Dep...	Enterprise Business Dept Director/Industr...	Delivery & Service Dept Director
	Account Manager	Solution Manager	D&S CSI and AMS engineers	Contract commercial manager
Specialist positions	Senior Partner Development Manager	CSPO	Quality Operations Dept Director	
	Basic Course Lecture Hall	Customer and partner Communication...	Vulnerability Management	Product and Contract Lifecycle Management
Scenario-based courses	Certificate management	Integrity Protection	Security configuration Management	

<https://ilearning.huawei.com/edx/subsite/page1/570621768656154624>

Expert Support Team: The CNBG HQ Expert Support Team Provides Enablement for Frontline Personnel and Engages in Transformation Projects

Transformation Project	Business Activity	CNBG Support Expert
General	/	Qiu Hong/00817786 (CNBG SPO) Yan Lin/00632056 (Quality & Operations)
Product and contract lifecycle management	Lifecycle clause incorporation in contracts	Zhang Hongxia/00643925 (Contract Commerce Dept)
	Lifecycle risk management	Mou Bilei/00387846 (GTS) Song Qingyan/00643779 (GTS)
	Exceptive service sales	Yu Chongjiu/00372172
	EOS software version incorporation	Zhang Nan/00473829 (MSSD)
Vulnerability management	Incorporation of vulnerability preventive clauses in contracts	Zhang Hongxia/00643925 (Contract Commerce Dept)
	Communication on high-risk vulnerabilities Routine communication via RNs	Qiu Hong/00817786 (CNBG SPO) Li Shiwei/00424696 (GSD) Huo Yonghong/00372054 (GTS)
	Risk mitigation after communication on high-risk vulnerabilities	Huo Yonghong/00372054 (GTS)
	Vulnerability handling	Huo Yonghong/00372054 (GTS)
Integrity protection	Tool-based signature verification solution (recommended) Direct connection to the software repository	Wang Peibing/00560947 (GTS) Xia Kaiwei/00398151 (GTS)
Certificate management	Risk mitigation for expired certificates on live networks	Wang Peibing/00560947 (GTS) Wang Lin/00820608
Security configuration management	Implementation of the basic configuration package Configuration inspection	Li Sheng'ai/00372280 (GTS) Huo Yonghong/00372054 (GTS)

Thank you

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